Tianxiang XIA

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EDUCATION

ETH Zürich, Zürich, Switzerland *since 09/2024*
Master's Degree, Major in Machine Intelligence GPA Major: **6/6** Overall: 5.87 (in progress)

Coursework Full marks (6/6) in Probabilistic AI, Deep Learning, Algorithms Lab, Optimizations for Data Science, Big Data, Research in CS Memo, etc.

Sorbonne Université, Paris, France *09/2021 - 06/2024*
Double Bachelor's Degree in Mathematics & Computer Science 225 ECTS Grade : **19/20 (rank 1/70)**

Coursework - Mathematics: topology and multivariate calculus, functional analysis, linear and bilinear algebra, measure theory and probability, numerical analysis, algebra and arithmetic, etc.
- Computer Science: AI and operational research, numerical methods, OOP and functional programming, databases, algorithms, data structures, logic, OS, cryptography, networks, computer architecture, etc.

AWARDS AND HONORS

- ETH Zürich Viscon-19 Hackathon, as part of a three-student team, **First Prize** *2024*
- International Collegiate Programming Contest, Southwestern Europe Regional Contest, **Bronze Medal** *2023*
- 30th China Team Selection Competition of International Olympiad in Informatics, **Silver Medal** *2018*
- National Olympiad of Informatics in Provinces of China, **First Prize** *2017*

SKILLS AND RESEARCH INTERESTS

Programming languages C/C++, Python, Java, OCaml, LaTeX, SQL, Intel/ARM Intrinsics

Spoken languages Chinese Mandarin (native), English (IELTS C1), French (C1, Bachelor's program in French)

Technical skills AI (PyTorch, vllm, LMCache), Unix-like Systems (Shell, Git, VIM), Docker, Web Development (HTML, CSS, JavaScript, React), Database Management (SQL, Spark, etc.)

Research Interests Optimizations for large models and systems, geometries of world-model latent variables, etc.

WORK EXPERIENCE

Research Student at ETH Zürich, SRI lab, ODI lab (present) *since 03/2025*
Teaching Assistant at Algorithms Lab (algolab), ETH Zürich *09/2025 - 12/2026, 09/2026 - 12/2026*

- In charge of oral assessment for students' performance in Master's level algorithms course

Research Intern at Computing Product Lab, Huawei Zürich *06/2025 - 08/2025*

- R&D of high-hit-rate KV cache systems (report accepted as DAC 2026 WIP session poster) and online quantization for multi-LoRA agents and retrieval-augmented generation scenarios.
- Contributed to open-source communities PTO-Kernels, LMCache-Ascend, CANN/ops-nn

Research Intern at Huawei Paris with report & code *06/2024 - 09/2024*

- Designed and built a neural-symbolic reasoning engine on automatic Euclidean geometry theorem proving with human-computer interaction features through reasoning graph plotting and GeoGebra integration
- The project was continued by HarmonicMath and achieved gold-medal performance at IMO 2025

PROJECTS

Learning Structured Sparsity for Large Models: Bridging Model Compressibility and Efficient Inference (submitted to ICLR 2027) *03/2026 - 09/2026*

- Developed an ADMM-based framework for jointly learning low-rank and hardware-friendly structured sparse representations during LLM pretraining
- Implemented an end-to-end pipeline with efficient low-rank proximal updates, a zero-shot elastic compression method, and custom TileLang/AVX2 inference kernels, cutting training overhead from 46% to 4% and advancing the empirical perplexity–memory–latency Pareto frontier

Hardware Optimization for Clifford Nets with Report & code, at ETH Zürich *03/2025 - 06/2025*

- Optimized Clifford neural layers with AVX2 and ARM NEON intrinsics, achieving a 20% speedup over PyTorch
- Developed a generator for vectorized C kernels across Clifford-algebra dimensions, signatures, and CPU architectures

Automatic Neural Network Robustness Verification with code, ETH Zürich *11/2024 - 12/2024*

- Implemented from scratch DeepPoly algorithms with learnable relaxations and obtained the highest evaluation score.